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Physics-Informed Neural Network for a Reduced WNT-HOX-Retinoic Acid Model of Colorectal Cancer

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neural net

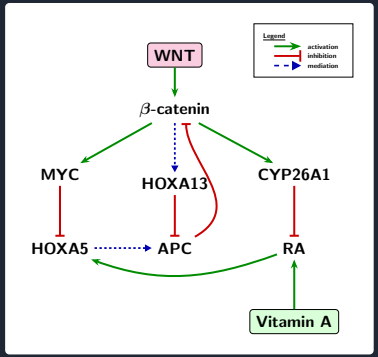
Overview

- Biology/Schematic
- Model
- Forward PINN
- Inverse PINN
- Inverse Bayesian PINN
- Next Steps

Biological Network Governing WNT–HOX–RA Signaling

Objective

Develop a reduced mathematical model describing the interactions among the **WNT**, **HOX**, and **Retinoic Acid (RA)** pathways during colorectal cancer progression.



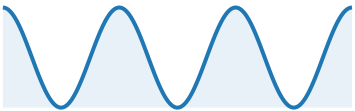
State Variables

$y = [B(t), P(t), H_5(t), H_{13}(t), M(t), R(t), C(t)]$

B: β-catenin
P: APC
H_5: HOXA5
H_{13}: HOXA13
M: MYC
R: Retinoic Acid
C: CYP26A1

RA Input + Treatment

RA Input



$$A_d \left[1 + \cos\left(\frac{2\pi t}{T_d} - \phi \right) \right]$$

ATRA Treatment



$$u_R(t) = u_0 + \frac{D}{2} \left[\tanh(q(t - t_1)) - \tanh(q(t - t_2)) \right].$$

Dimensional Model

$$\frac{dB}{dt} = a_B W + v_{13}\frac{H_{13}^{n_H}}{K_{13}^{n_H} + H_{13}^{n_H}} - d_B B - k_{PB} P B - v_5 \frac{H_5 B}{K_5 + B}$$

$$\frac{dP}{dt} = a_P \frac{1 + \rho_5 H_5}{1 + \rho_B B + \rho_{13} H_{13}} - \Big(d_{P0} + d_{P1}(1 - \Theta_P)\Big) P$$

$$\frac{dH_5}{dt} = a_5 + v_R \frac{R}{K_R + R} - d_5 H_5 - v_M \frac{M H_5}{K_M + M}$$

$$\frac{dH_{13}}{dt} = a_{13} + v_{B13}\frac{B^{n_B}}{K_{B13}^{n_B} + B^{n_B}} + v_{M13}\frac{M^{n_M}}{K_{M13}^{n_M} + M^{n_M}} - d_{13} H_{13}$$

$$\frac{dM}{dt} = a_M + v_{BM}\frac{B^{n_B}}{K_{BM}^{n_B} + B^{n_B}} - d_M M$$

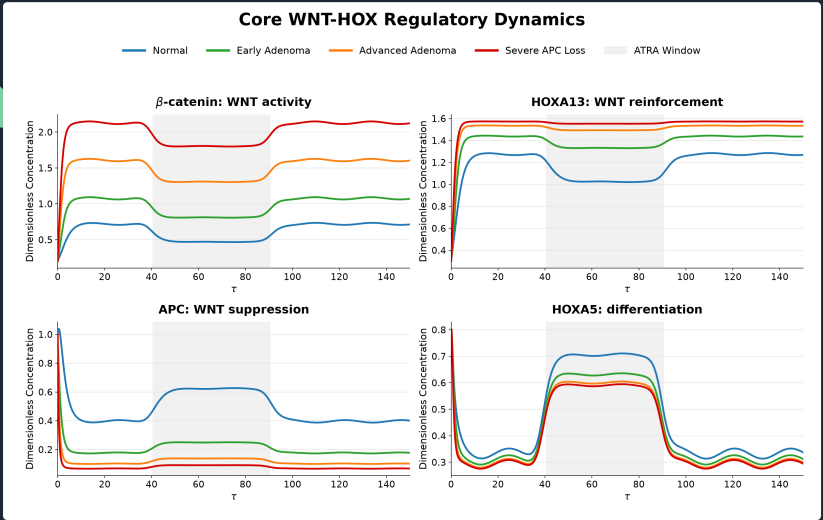
$$\frac{dR}{dt} = u_0 + A_d\left[1 + \cos\left(\frac{2\pi t}{T_d} - \phi\right)\right] + \frac{D}{2}\Big[\tanh(q(t - t_1)) - \tanh(q(t - t_2))\Big] - d_R R - k_{CR} C R$$

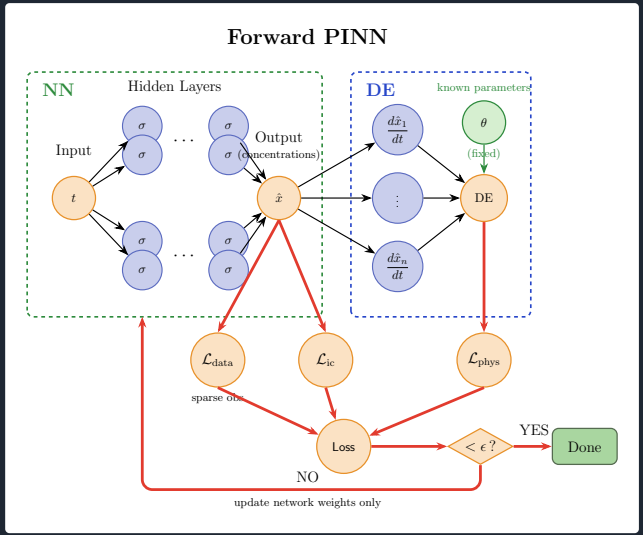
$$\frac{dC}{dt} = a_C + v_{RC}\frac{R}{K_{RC} + R} + v_{BC}\frac{B^{n_B}}{K_{BC}^{n_B} + B^{n_B}} - d_C C$$

4 Regimes

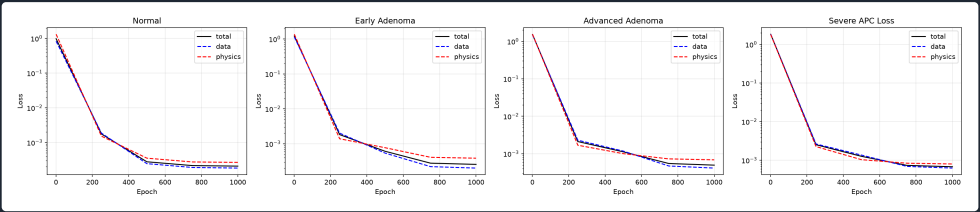
Disease progression is modeled by increasing WNT signaling (W) and progressive loss of APC function (Θ_p), mimicking CRC progression.

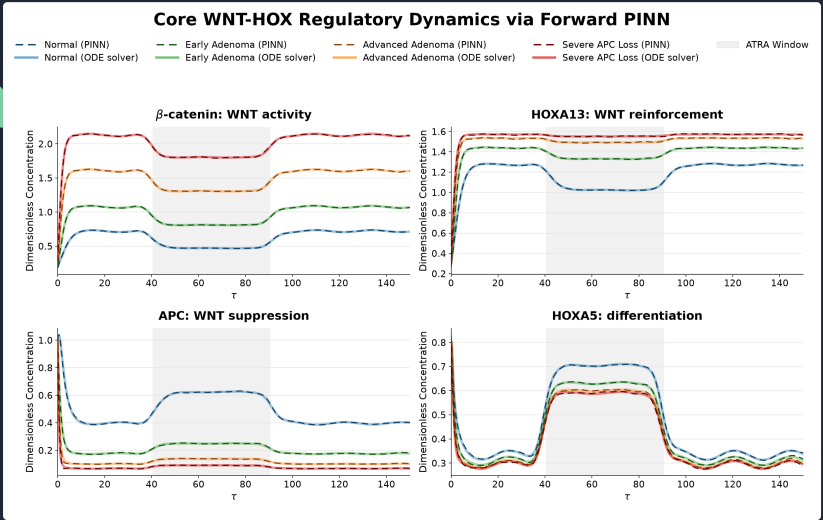
- “Normal”
  - $W = \mathbf{0.8}, \quad \Theta_p = \mathbf{1.0}$
- “Early Adenoma”
  - $W = \mathbf{1.0}, \quad \Theta_p = \mathbf{0.75}$
- “Advanced Adenoma”
  - $W = \mathbf{1.5}, \quad \Theta_p = \mathbf{0.5}$
- “Severe APC Loss”
  - $W = \mathbf{2.0}, \quad \Theta_p = \mathbf{0.25}$

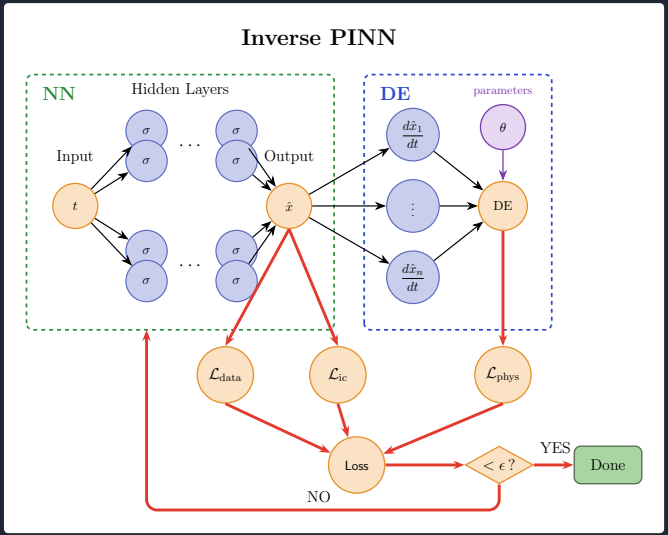

Core WNT-HOX Regulatory Dynamics
Normal
Early Adenoma
Advanced Adenoma
Severe APC Loss
ATRA Window
β-catenin: WNT activity
HOXA13: WNT reinforcement
APC: WNT suppression
HOXA5: differentiation
Dimensionless Concentration
τ


Forward PINN
NN
Hidden Layers
Input
Output
(concentrations)
DE
known parameters
(fixed)
DE
sparse obs.
Loss
YES
Done
NO
update network weights only

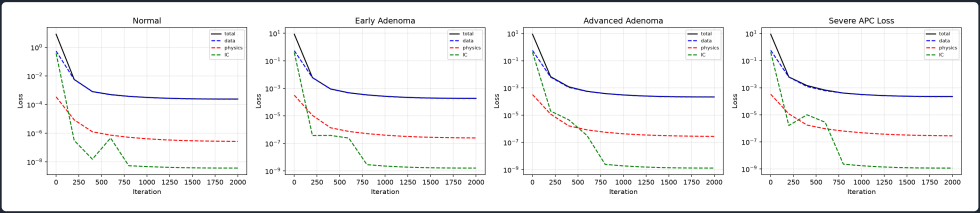
ward PINN Loss

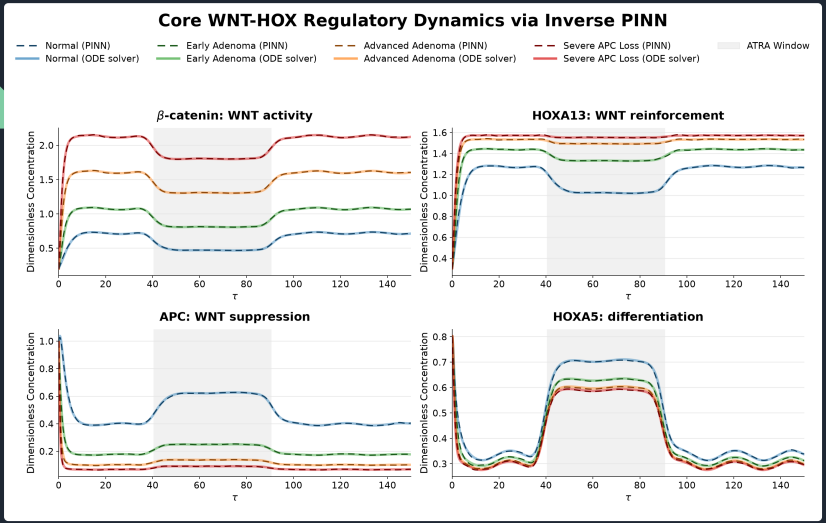


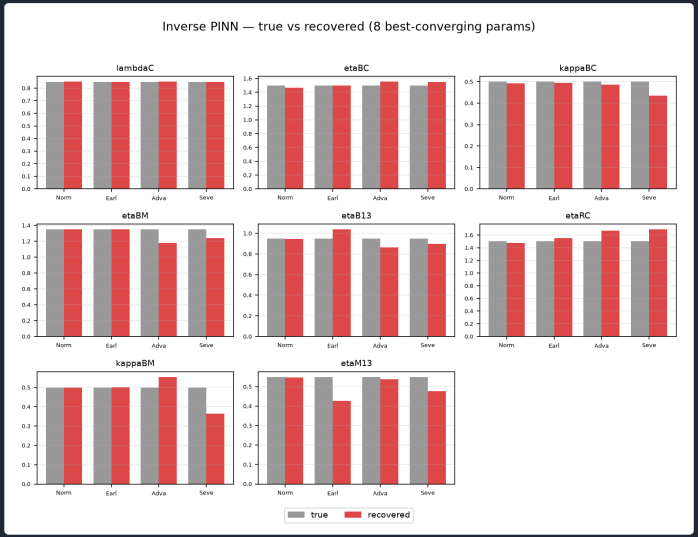

Core WNT-HOX Regulatory Dynamics via Forward PINN
Normal (PINN)
Normal (ODE solver)
Early Adenoma (PINN)
Early Adenoma (ODE solver)
Advanced Adenoma (PINN)
Advanced Adenoma (ODE solver)
Severe APC Loss (PINN)
Severe APC Loss (ODE solver)
ATRA Window
β-catenin: WNT activity
HOXA13: WNT reinforcement
APC: WNT suppression
HOXA5: differentiation
Dimensionless Concentration
τ


Inverse PINN
NN
Hidden Layers
Input
Output
t
σ
$\hat{x}$
DE
parameters
$\frac{d\hat{x}_1}{dt}$
$\frac{d\hat{x}_n}{dt}$
θ
DE
$\mathcal{L}_{\text{data}}$
$\mathcal{L}_{\text{ic}}$
$\mathcal{L}_{\text{phys}}$
Loss
$< \epsilon$?
YES
NO
Done

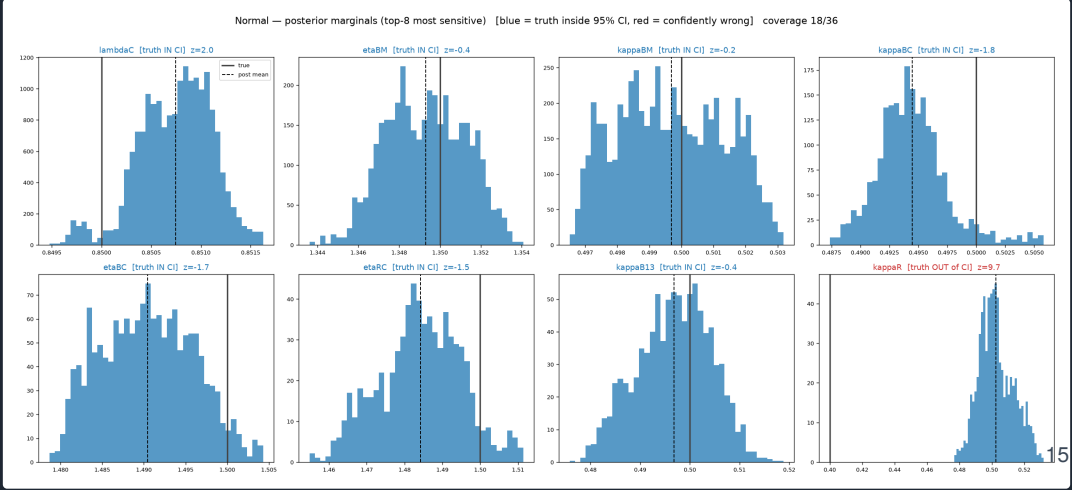
Inverse PINN Loss



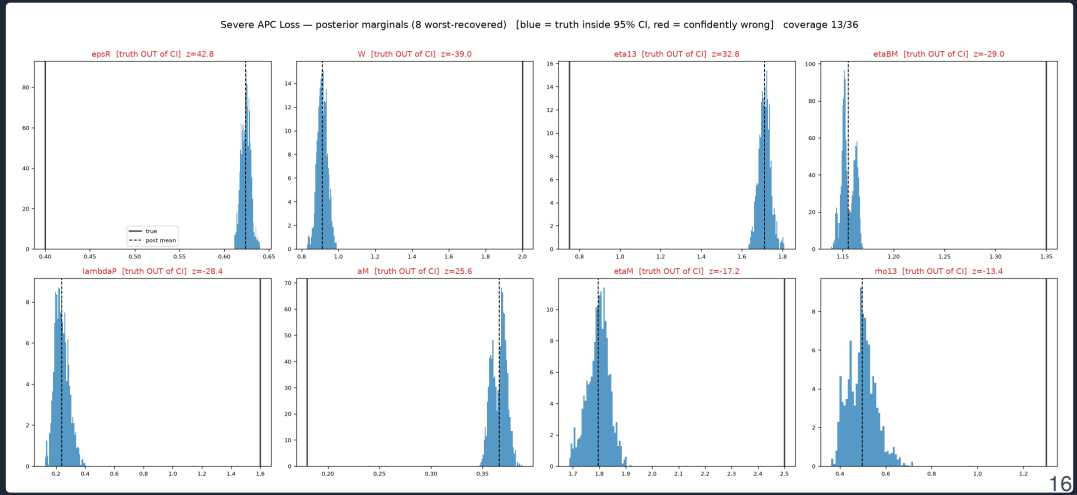

Core WNT-HOX Regulatory Dynamics via Inverse PINN
Normal (PINN)
Normal (ODE solver)
Early Adenoma (PINN)
Early Adenoma (ODE solver)
Advanced Adenoma (PINN)
Advanced Adenoma (ODE solver)
Severe APC Loss (PINN)
Severe APC Loss (ODE solver)
ATRA Window
β-catenin: WNT activity
HOXA13: WNT reinforcement
APC: WNT suppression
HOXA5: differentiation
Dimensionless Concentration
τ


Inverse PINN — true vs recovered (8 best-converging params)
lambdaC
etaBC
kappaBC
etaBM
etaB13
etaRC
kappaBM
etaM13
Norm
Earl
Adva
Seve
true
recovered

Inverse Bayesian PINN — uncertainty quantification



Inverse Bayesian PINN — continued



Parameter Recovery Comparison

	Bayesian inverse PINN	Inverse PINN	ODE-fit
“Normal”	**19**	**17**	**18/36**
“Early Adenoma”	**20**	**16**	**17/36**
“Advanced Ade-	**11**	**10**	**13/36**

Next Steps

- Improve Bayesian PINN
  - Investigate limitations
- Hybrid Mechanistic-Neural Model
- Stochastic Modeling

Next Steps graphic *missing*

Key References

1. **Raissi, Perdikaris & Karniadakis (2019).** Physics-informed neural networks: a deep learning framework for solving forward and inverse problems involving nonlinear PDEs. *J. Comput. Phys.* **378**, 686–707.

   Foundational PINN formulation — the data + physics-residual loss used in every model here.

2. **Yazdani, Lu, Raissi & Karniadakis (2020).** Systems biology informed deep learning for inferring parameters and hidden dynamics. *PLoS Comput. Biol.* **16**(11), e1007575.

   Closest template to this work: inverse PINNs on biological ODE networks; the local-sensitivity/Fisher route to identifiability.

3. **Yang, Meng & Karniadakis (2021).** B-PINNs: Bayesian physics-informed neural networks for forward and inverse PDE problems with noisy data. *J. Comput. Phys.* **425**, 109913.

   Basis of the Bayesian inverse PINN: HMC over parameters gives posterior marginals instead of a point estimate.

4. **Raue *et al.* (2009).** Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics* **25**(15), 1923–1929.

Thank you!

Questions?

Thank you very much to the Provost's Office for funding our summer research project!